Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

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Title: Intravenous immunoglobulins (IVIG) replacement guideline for children with primary immunodeficiency

Introduction:

- Immunoglobulins are glycoprotein molecules produced by plasma cells in response to a variety of antigenic stimuli and are involved in diverse physiological and pathological processes. Immunoglobulins (natural antibodies bridge innate and adaptive Immunity) primarily function in the adaptive arm of the immune system and are subdivided, based ontheir heavy chains, into various classes, i.e., IgM, IgG, IgD, IgA, and IgE. IgG is the most abundant immunoglobulin within the plasma.
- What is IVIG? It is a manufactured mixture of IgG antibodies against a wide spectrum of infectious pathogensderived from plasma of thousands of healthy donors. IVIG may contains traces of IgM and IgA.

- What are the conditions that IVIG is used for? There are many clinical uses of IVIG, however it is used in various primary immunodeficiency syndromes especially those with immunoglobulin defects/deficiency. Examples are:x-linked agammaglobulinemia, autosomal recessive agammaglobulinemia, severe combined and combined immunodeficiencies, hyper IgM syndrome, common variable immunodeficiency, transient hypogammaglobulinemia of infancy and others.
- What is the basic goal from regular IVIG administration? To Reduce the incidence and the severity of infections in patients with immunodeficiency, decrease hospitalization and antibiotic usage and improve pulmonary outcomes of these patients.

Dose & Administration.

- Is it a routine to administer pre-medications for IVIG infusion? No, pre-medications administration is not a routine, and it should only be reserved for children who experienced mild or moderate side effects from previous IVIG infusions.
- IVIG is administered intravenously as an infusion **over 6 hours at initial infusion and over 4 hours upon subsequent infusions. Subcutaneous immunoglobulin is another option of administration however not widely available in our region.**
- Replacement dose is 400 to 600 mg/kg every 3 – 4 weeks.Please titrate to the nearest high final volume according to the vial used (2.5- or 5-gram/vial) to minimize waist.
- IgG trough levels (defined as the lowest concentration achieved before the next dose) of 500 to 800 mg/dL are achievable with this dosing and considered to be protective from infectious consequences in immunodeficient patients. We should aim for an IgG trough level of 800 mg/dL and above in children suffering from chronic lung diseases like bronchiectasis.
- Please contact the immunology team if the desired trough level is not achieved after proper adjustments.

Adverse Effects

- Adverse reactions can be either immediate or delayed and can be of different severity.

- **Immediate**

1. **Mild Reactions**

- **Itching**
- **Muscle aches**
- **Flushing**
- **Mild headache**
- **Shivering/ low grade fever**
- **Nausea**

- These reactions may not require the infusion to be stopped.
- The rate should be slowed to its minimum.
- Oral paracetamol (15mg/kg) and chlorpheniramine (0.1 mg/kg) to be given when symptomatic.
- If symptoms subside, the infusion rate may be gradually increased.

2. Moderate reactions

- **Itchy, urticarial rash**
- **Severe headache**
- **Worsening or re-occurrence of mild reaction**

- The infusion should be stopped and oral paracetamol (15mg/kg), oral or intravenous chlorpheniramine (0.1 mg/kg) and IV hydrocortisone (2 mg/kg) to be given.
- The patient should be closely observed and monitored for the worsening of symptoms.
- Can use salbutamol nebulizer/puff for bronchospasm.
- The infusion may be recommenced slowly (over 12 hours) and cautiously if symptoms are relieved.

3. Severe Reaction (very rare but could potentially occur)

- **Chest pain or pressure in the chest.**
- **Tightness of the throat.**
- **Severe dizziness or fainting.**
- **Bronchospasm/wheeze.**
- **Collapse/severe hypotension.**
- **Moderate symptoms reoccurring or becoming worse.**

- In this instance **STOP** the infusion.
- Lay the patient flat and raise the legs. Check the pulse and auscultate the chest.
- InjectIM adrenaline (0.01 mg/kg) into the antero-lateral aspect of the thigh.
- Activation of emergency response.
- Administer the medications for moderate symptoms after the first dose of IM adrenaline and if wheezing, to use salbutamol nebulizer/puff.
- Continuous monitoring.
- Update the pediatric immunology team for further advise.
- Other rare severe reactions include hemolysis, thromboembolic events, and aseptic meningitis syndrome.

- Less than 1% of patients may have a delayed reaction, which includes renal impairment, transfusion-related infection, hematological, and neurological disorders.
- Please discuss with pediatric immunology team for further advise about these adverse reactions.

Monitoring

- Patients receiving initial (first-time) IVIG infusions should be monitored carefully for any infusion-related reactions or adverse effects.

- Standard recordings prior to the first infusion & regularly monitored.

- IgG trough level (repeat 3 monthly)
- Liver enzymes (repeat 6 monthly)
- Full blood count (repeat 6 monthly)

- Body weight (repeat 6 monthly)
- Hepatitis B &hepatitis C, CMV, HIV PCRs (repeat every 1 – 3 years). The risk of transmission of infectious pathogens by IVIG is minimized by the careful selection of donors, plasma antibody screening, and various procedures of viral inactivation.

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